

Assignment No.

03

Combinational

Logic Design

FAROOQ ABDULLA

Online Internship on Analog and Digital VLSI Design Flow

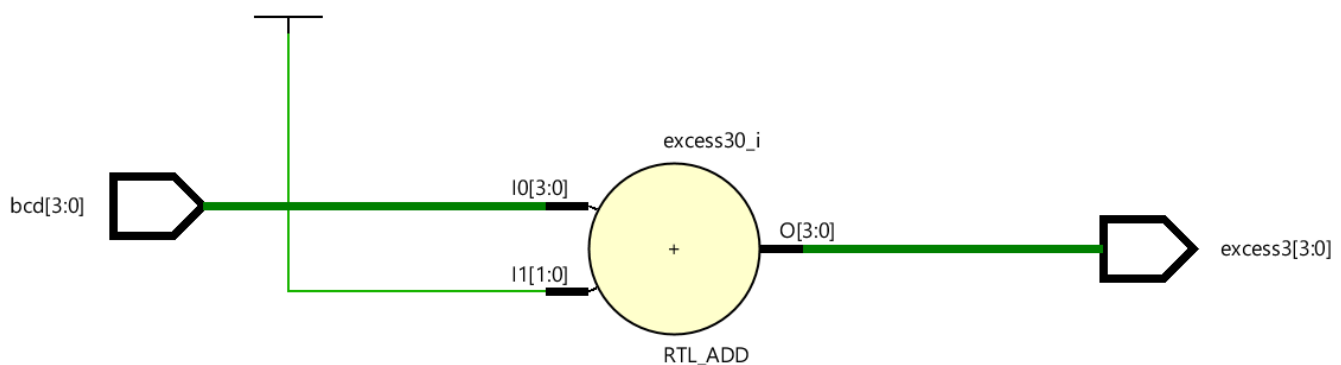
Batch 01

Q1. Design and Simulate 4-bit BCD to Excess-3 code converter using Verilog HDL and verify its operation through simulation.

DESIGN CODE:

```
module bcd_to_excess3 (  
    input [3:0] bcd,  
    output reg [3:0] excess3);  
  
    always @(*) begin  
        if (bcd <= 4'd9) begin  
            excess3 = bcd + 4'b0011;  
        end else begin  
            excess3 = 4'bxxxx;  
        end  
    end  
  
endmodule
```

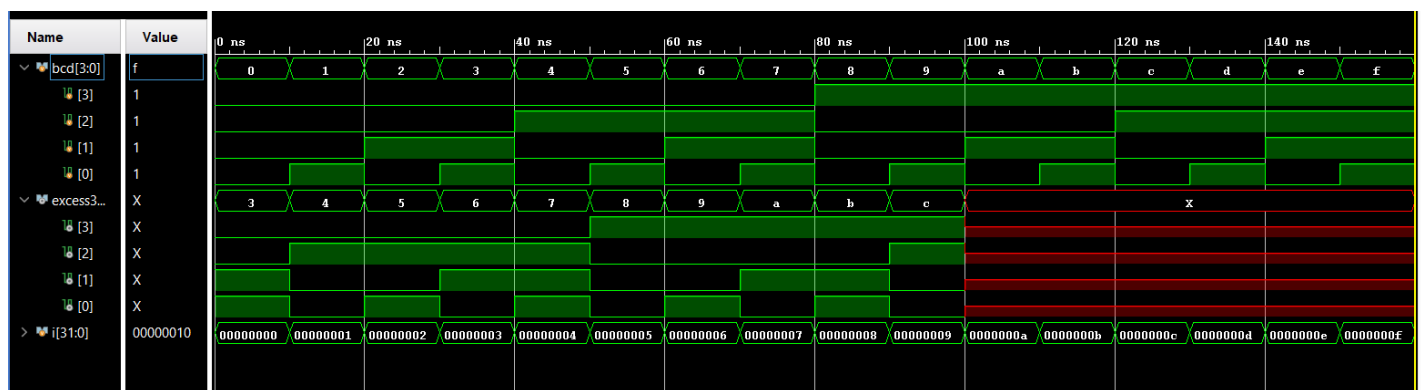
SCHEMATIC:



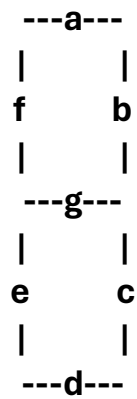
TESTBENCH CODE :

```
module tb_bcd_to_excess3;  
    reg [3:0] bcd;  
    wire [3:0] excess3;  
  
    integer i;  
  
    bcd_to_excess3 uut (.bcd(bcd),.excess3(excess3));  
  
    initial begin  
        $display("Time\t BCD Input \t Excess-3 Output");  
        $monitor("%0t\t    %b (%d) \t    %b", $time, bcd, bcd, excess3);  
  
        for (i = 0; i < 16; i = i + 1) begin  
            bcd = i;  
            #10;  
        end  
  
        $finish;  
    end  
endmodule
```

SIMULATION:



Q2. Design and simulate a Verilog module that converts a 4-bit BCD input into the corresponding seven segment display outputs.

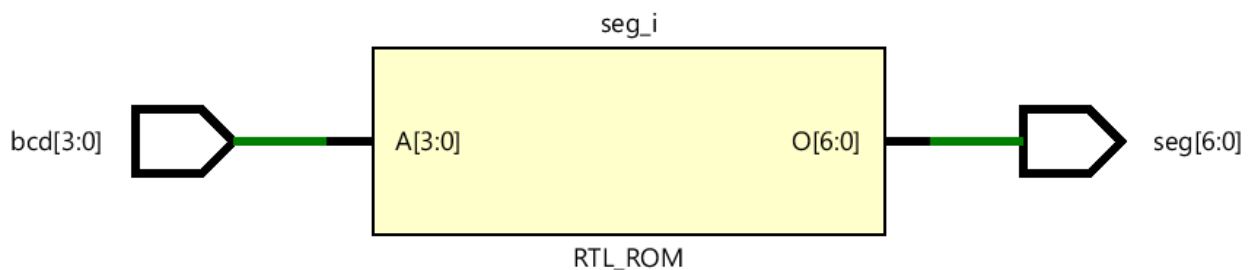


BCD	Segment Output (abcdefg)	Hex Display
0000	1111110	0
0001	0110000	1
0010	1101101	2
0011	1111001	3
0100	0110011	4
0101	1011011	5
0110	1011111	6
0111	1110000	7
1000	1111111	8
1001	1111011	9

DESIGN CODE:

```
module bcd_to_seven_seg (  
    input [3:0] bcd,  
    output reg [6:0] seg );  
  
    always @(*) begin  
        case (bcd)  
            4'b0000: seg = 7'b1111110; // 0  
            4'b0001: seg = 7'b0110000; // 1  
            4'b0010: seg = 7'b1101101; // 2  
            4'b0011: seg = 7'b1111001; // 3  
            4'b0100: seg = 7'b0110011; // 4  
            4'b0101: seg = 7'b1011011; // 5  
            4'b0110: seg = 7'b1011111; // 6  
            4'b0111: seg = 7'b1110000; // 7  
            4'b1000: seg = 7'b1111111; // 8  
            4'b1001: seg = 7'b1111011; // 9  
            default: seg = 7'b0000000;  
        endcase  
    end  
endmodule
```

SCHEMATIC:



TESTBENCH CODE:

```
module tb_bcd_to_seven_seg;
    reg [3:0] bcd;
    wire [6:0] seg;

    integer i;

    bcd_to_seven_seg uut (
        .bcd(bcd),
        .seg(seg));

    initial begin
        $display("Time\t BCD Input \t Segment Output (abcdefg)");
        $monitor("%0t\t    %b (%d) \t          %b", $time, bcd, bcd, seg);

        for (i = 0; i < 16; i = i + 1) begin
            bcd = i;
            #10;
        end

        $finish;
    end
endmodule
```

SIMULATION:

